

Manufacturing Quality Dashboard

10K
Total Machines

339
Total Failures

3.39%
Failure_Rate

348
Failure_Modes

27
Label Mismatch

ProductType

☐ H

☐ L

☐ M

FailureMode

☐ HDF

☐ OSF

☐ PWF

WearBucket

☐ 0-50

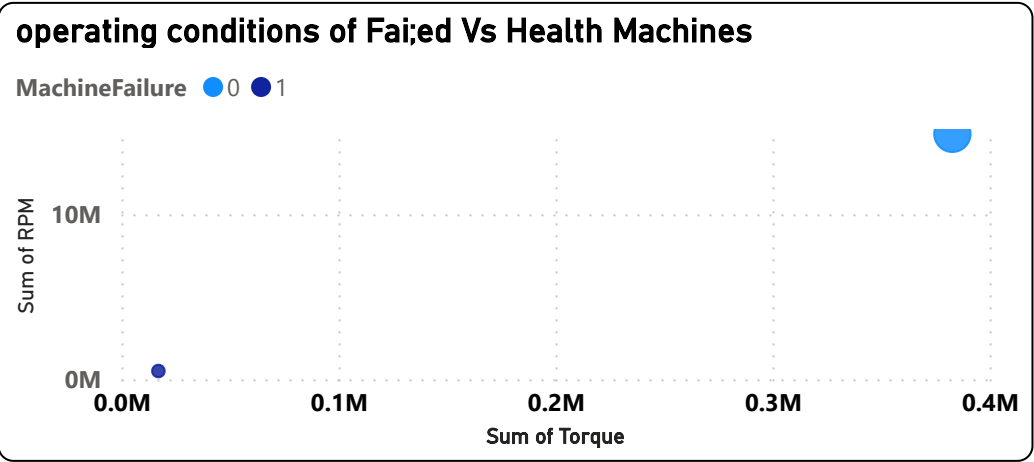
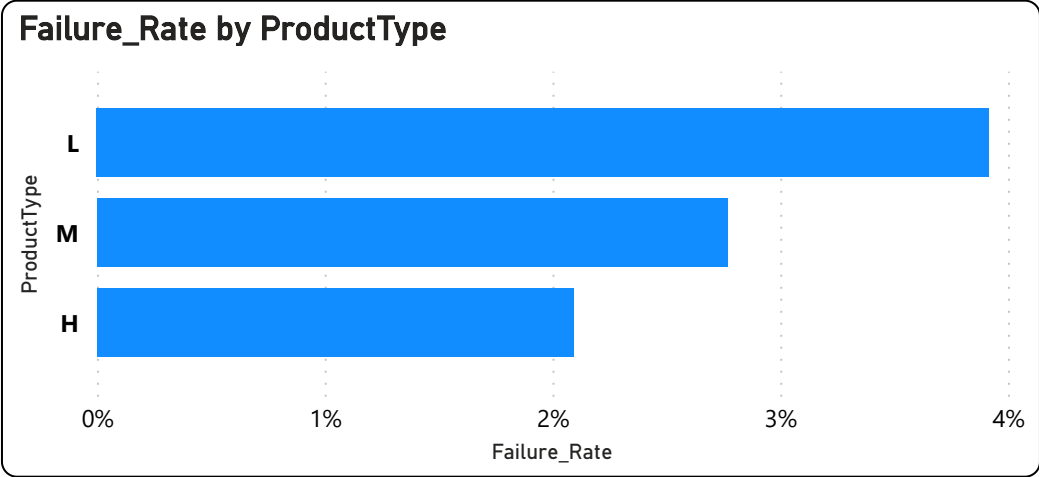
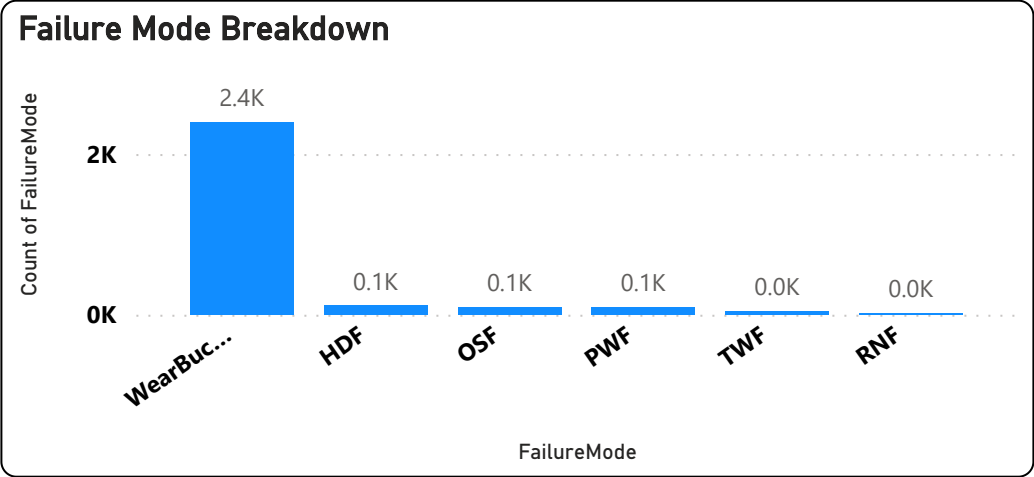
☐ 51-100

☐ 101-150

Machine Failure Stat...

☐ Failed

☐ No Failure



Executive Summary

Failure rate is 3.39%. Heat Dissipation Failure is the most common failure mode. Machines operating with high torque, elevated temperature gap, and excessive tool wear show the highest f...



Operating Conditions

ProductType

☐ H

☐ L

☐ M

FailureMode

☐ HDF

☐ OSF

WearBucket

☐ 0-50

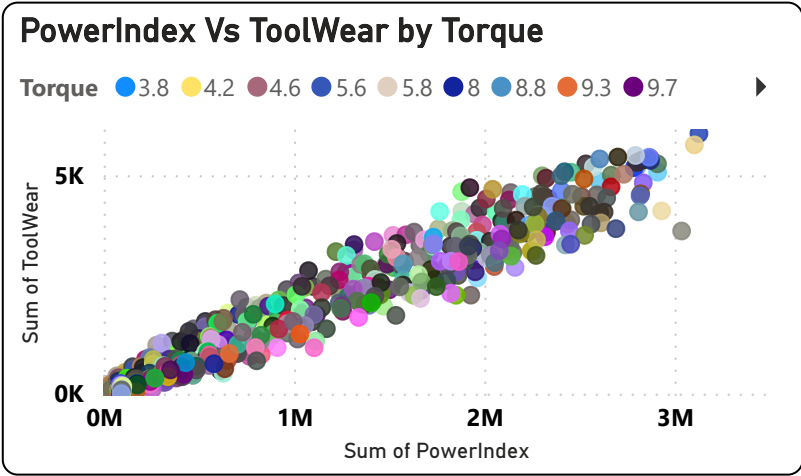
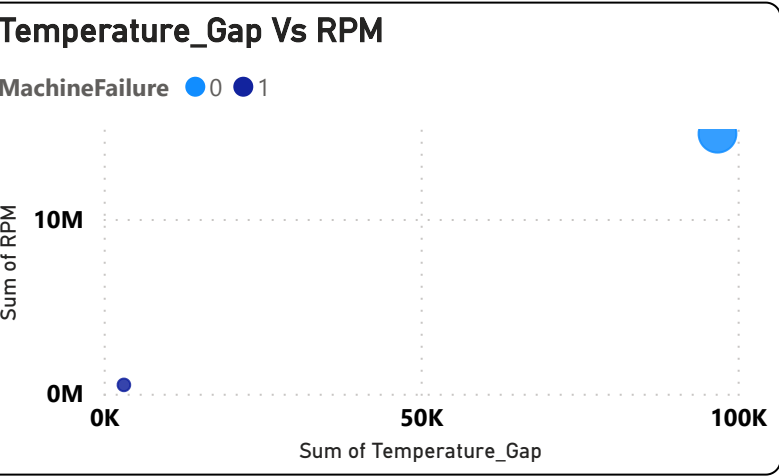
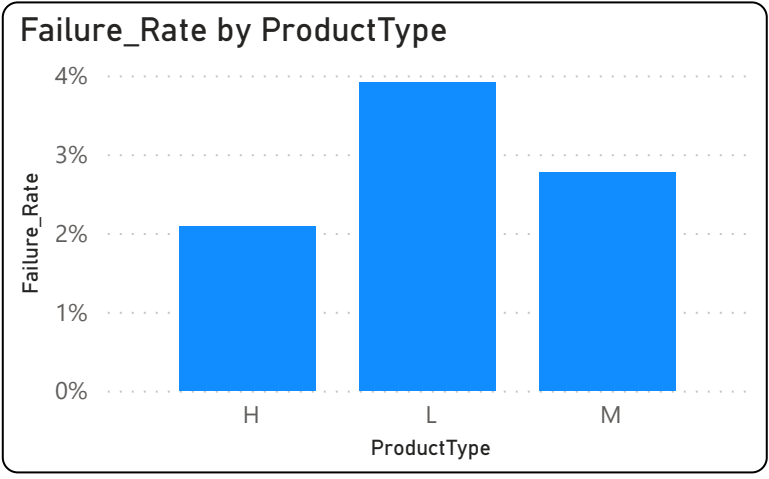
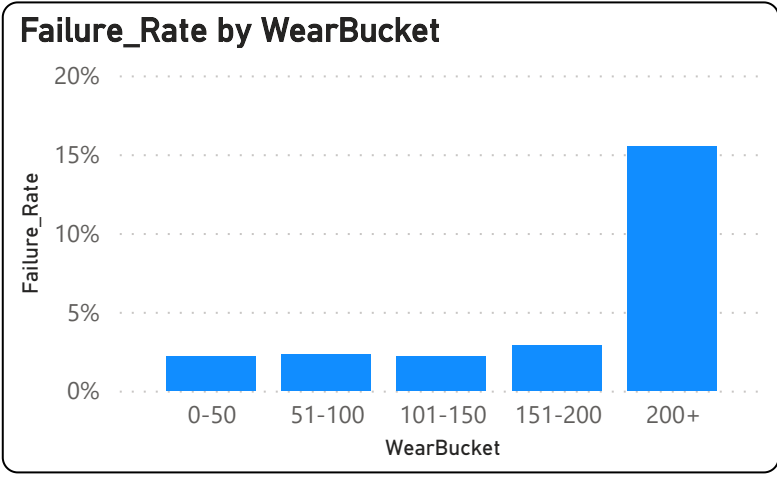
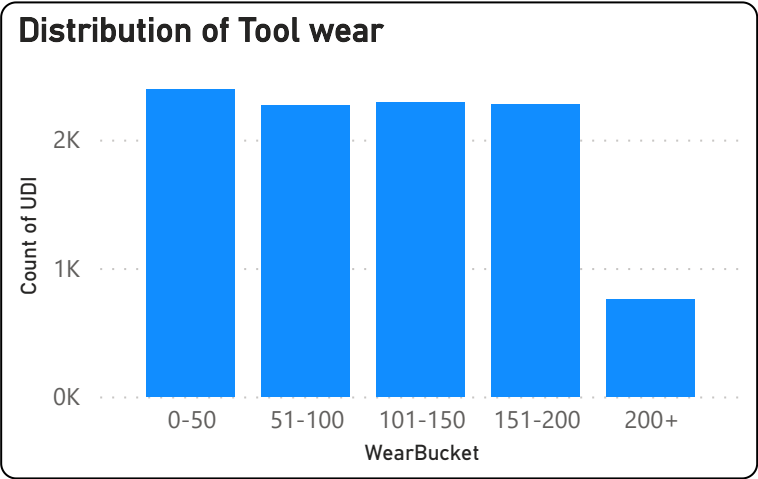
☐ 51-100

Machine Failure

☐ Failed

☐ No Failure

Machine Failure



Wear Bucket vs Failure Rate

WearBucket	Total_Machines	Sum of MachineFailure	Failure_F
0-50	2396	52	2.1
101-150	2295	50	2.1
151-200	2276	66	2.9
51-100	2271	53	2.3
200+	762	118	15.4
Total	10000	339	3.3

Failure Mode Analysis

ProductType

- ☐ H
- ☐ L
- ☐ M

FailureMode

- ☐ HDF
- ☐ OSF
- ☐ PWF

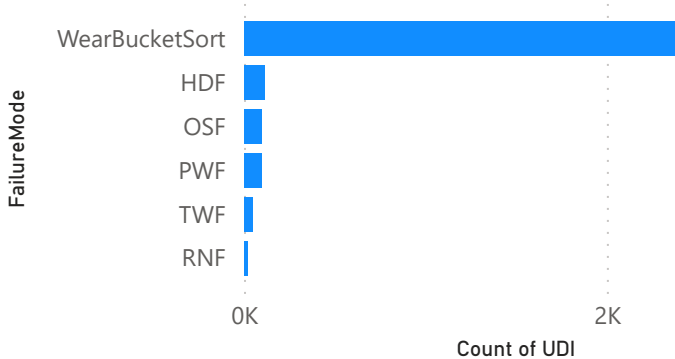
WearBucket

- ☐ 0-50
- ☐ 51-100
- ☐ 101-150

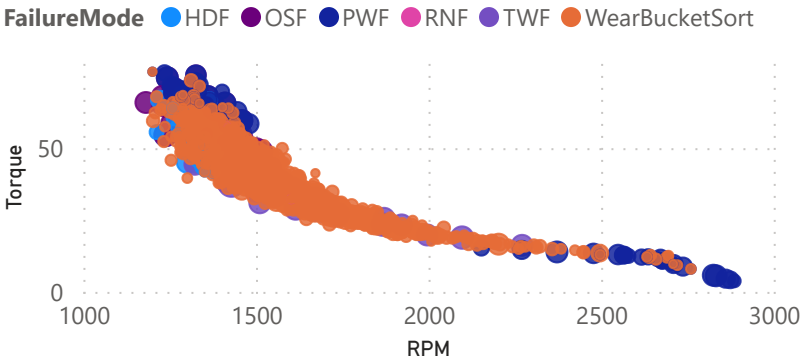
Machine Failure Stat...

- ☐ Failed
- ☐ No Failure

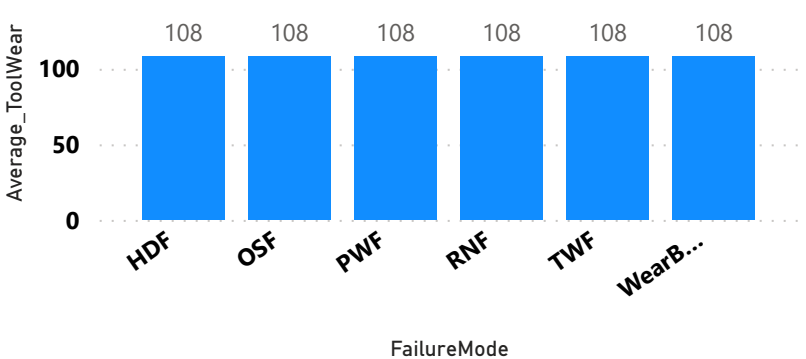
Failure Mode Breakdown



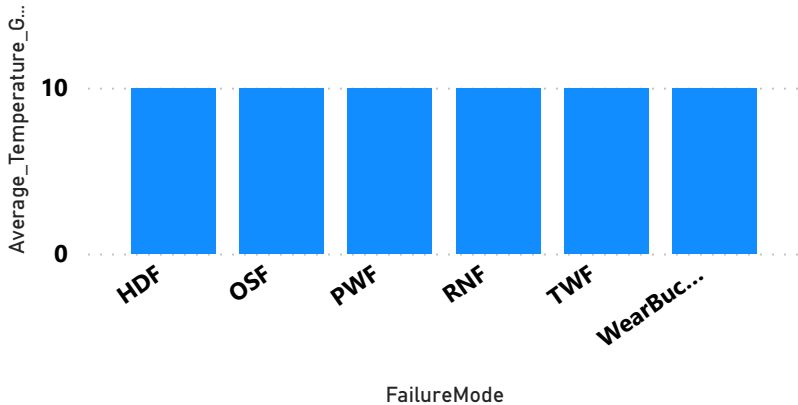
RPM vs Torque Scatter Plot



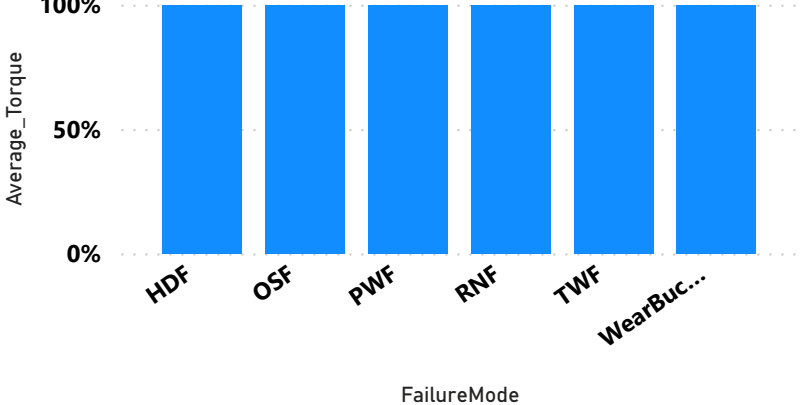
Average_ToolWear by FailureMode



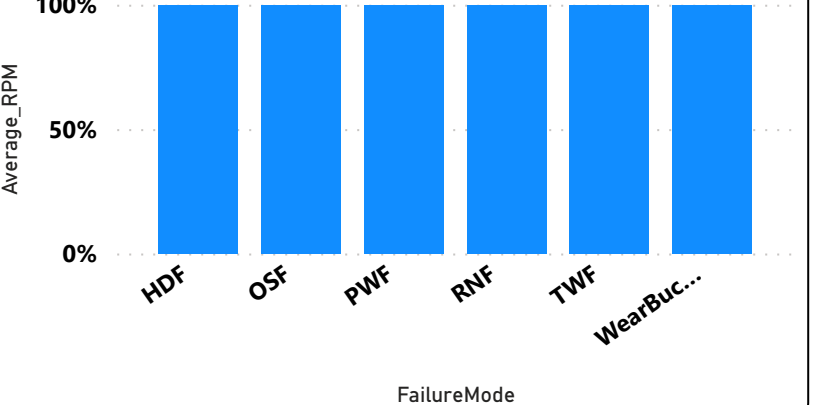
Average_Temperature_Gap by FailureMode



Average_Torque by FailureMode



Average_RPM by FailureMode



Engineer Monitoring Dashboard

Temperature_Gap

10.00

Average_Temp...

RPM

1.54K

Average_RPM

Torque

40.00

Average_Torque

TotalWear

108.38

Average_Tool...

ProductType ▾

☐ H

☒ L

☐ M

FailureMode ▾

☐ HDF

☐ OSF

☐ PWF

☒ RNF

WearBucket ▾

☐ 0-50

☐ 51-100

☐ 101-150

☐ 151-200

Machine Failure Stat... ▾

☐ Failed

☐ No Failure

Monitoring Matrix

Condiiton	First Normal	First Watch	First Danger
RPM	✓		
Temperature_Gap	✓		
ToolWear		✓	
Torque			✓
Total			

Engineering Recommendations

- Monitor Temperature Gap continuously.
- Replace tools before Tool Wear exceeds 180–200 minutes.
- Machines operating with high Torque should be inspected.
- Investigate machines with high Temperature Gap and low RPM.
- Prioritize HDF and OSF machines for preventive maintenance.

Engineering Recommendations

- Temperature gap acc...
Temperature Recomme...
- Tool wear within acc...
Wear Recommendation
- RPM within operati...
RPM Recommendation
- Torque operating n...

Data Quality Dashboard

ProductType

- ☐ H
☐ L
☐ M

FailureMode

- ☐ HDF
☐ OSF
☐ PWF

WearBucket

- ☐ 0-50
☐ 51-100
☐ 101-150

Machine Failure Stat...

- ☐ Failed
☐ No Failure

27

Label Mismatch

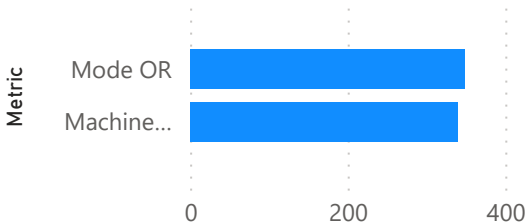
0

Missing Records

348

Sum of ModeOR

Comparison Value by Metric



Data Quality Summary

Check	Result
Label Mismatch	27
Duplicate Rows	None
Missing Values	None
Mode OR Check	Passed with exceptions

Data Quality Assessment

- No missing values were identified in the dataset.
- A small number of records (approximately 27) have a mismatch between the Machine Failure flag and the combined failure mode (Mode OR).
- These mismatches were retained for transparency and highlighted in the dashboard rather than removed.
- Overall, the dataset is suitable for exploratory analysis, but these inconsistencies should be reviewed with the data owner before using it for production reporting or predictive models.

Assessment

Dataset is suitable for analysis.

27 label mismatches retained for transparency.

Machine Failure Dashboard

Temperature_Gap

10.00

Average_Temp...

RPM

1.54K

Average_RPM

Torque

39.99

Average_Torque

TotalWear

107.95

Average_Tool...

ProductType

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- ☐ L
- ☐ M

FailureMode

- ☐ HDF
- ☐ OSF
- ☐ PWF

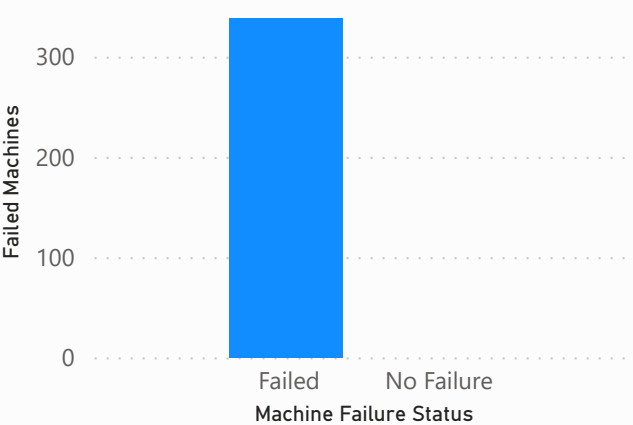
WearBucket

- ☐ 0-50
- ☐ 51-100
- ☐ 101-150

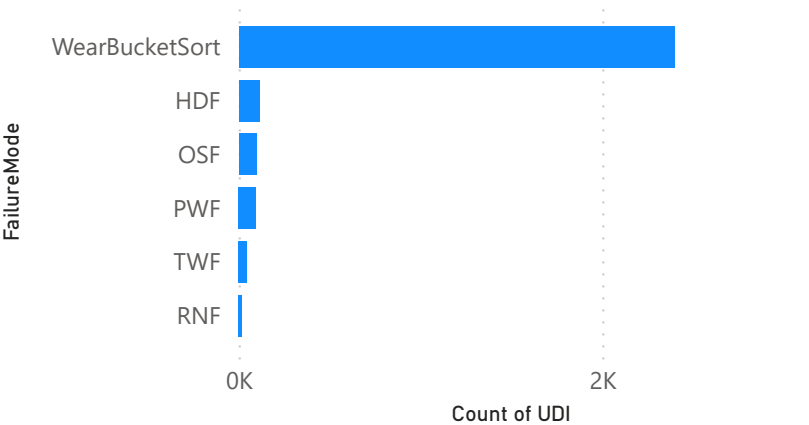
Machine Failure Stat...

- ☐ Failed
- ☐ No Failure

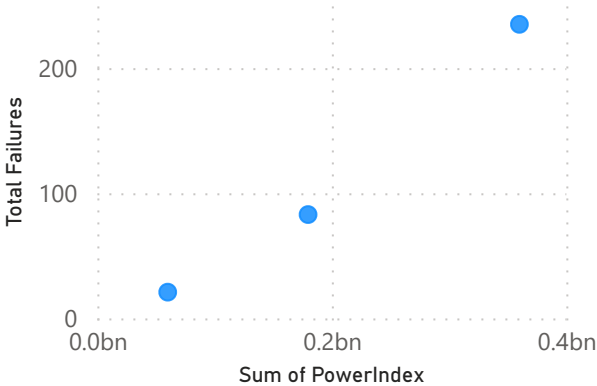
Failed Machines by Machine Failure Status



Failure Mode Breakdown



Sum of PowerIndex and Total Failures by ProductType



Monitoring Matrix

Condiiton	First Normal	First Watch	First Danger
RPM	✓		
Temperature_Gap	✓		
ToolWear		✓	
Torque			✓
Total			

Executive Summary by Failure Rate

Failure rate is 3.39%. 348 is the most common failure mode. Machines operating with hig...

Summary

ProductType

☐ H
☐ L
☐ M

FailureMode

☐ HDF
☐ OSF
☐ PWF

WearBucket

☐ 0-50
☐ 51-100
☐ 101-150

Machine Failure Status

☐ Failed
☐ No Failure

3.39%

Failure_Rate

348

Failure_Modes

339

Failed Machines

27

Label Mismatch

Executive Sum...

Failure rate is 3.39%. Heat Dissipation Failure is the most common failure mode. Machines

Executive Summary

Failure rate is 3.39%. Heat Dissipation Failure is the most common failure mode. Machines operating with high torque, elevated temperature gap, and excessive tool wear show the highest failure risk. Engineers should monitor these operating conditions for preventiv...

Executive Summary

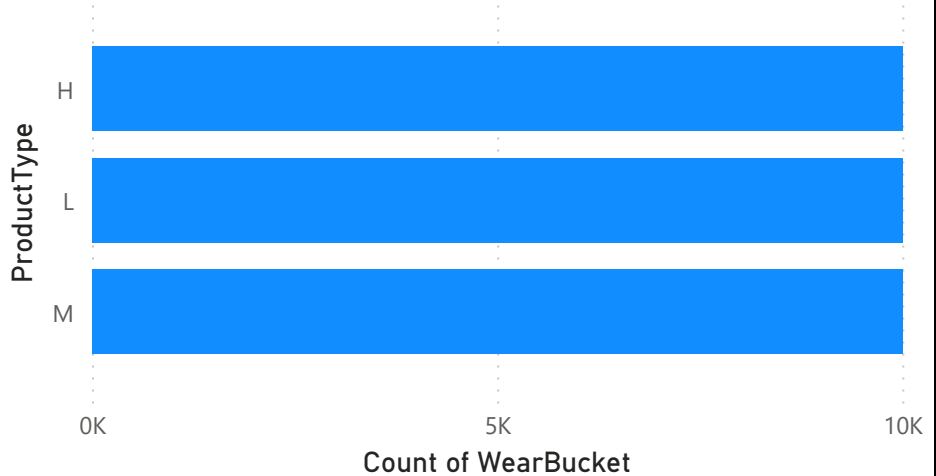
Executive Summary by Failure Rate

Failure rate is 3.39%. 348 is the most common failure mode. Machines operating with hig...



Drill through for details

Count of WearBucket by ProductType



⏪

ProductType

⬆

ProductType	ProductKey	Description	UDI	Failure_Modes	Failure_Rate	Failed Machines
L	1	Low Quality	2	0	0.00%	0
L	1	Low Quality	3	0	0.00%	0
L	1	Low Quality	4	0	0.00%	0
L	1	Low Quality	5	0	0.00%	0
L	1	Low Quality	7	0	0.00%	0
L	1	Low Quality	8	0	0.00%	0
L	1	Low Quality	15	0	0.00%	0
L	1	Low Quality	16	0	0.00%	0
L	1	Low Quality	22	0	0.00%	0
L	1	Low Quality	24	0	0.00%	0
L	1	Low Quality	26	0	0.00%	0
L	1	Low Quality	27	0	0.00%	0
L	1	Low Quality	29	0	0.00%	0
L	1	Low Quality	30	0	0.00%	0
L	1	Low Quality	32	0	0.00%	0
L	1	Low Quality	33	0	0.00%	0
L	1	Low Quality	34	0	0.00%	0
L	1	Low Quality	38	0	0.00%	0
L	1	Low Quality	40	0	0.00%	0
L	1	Low Quality	41	0	0.00%	0
L	1	Low Quality	42	0	0.00%	0
L	1	Low Quality	46	0	0.00%	0
L	1	Low Quality	48	0	0.00%	0
L	1	Low Quality	51	1	100.00%	1
L	1	Low Quality	52	0	0.00%	0
L	1	Low Quality	54	0	0.00%	0
L	1	Low Quality	55	0	0.00%	0
L	1	Low Quality	56	0	0.00%	0
L	1	Low Quality	57	0	0.00%	0
Total				348	3.39%	339

